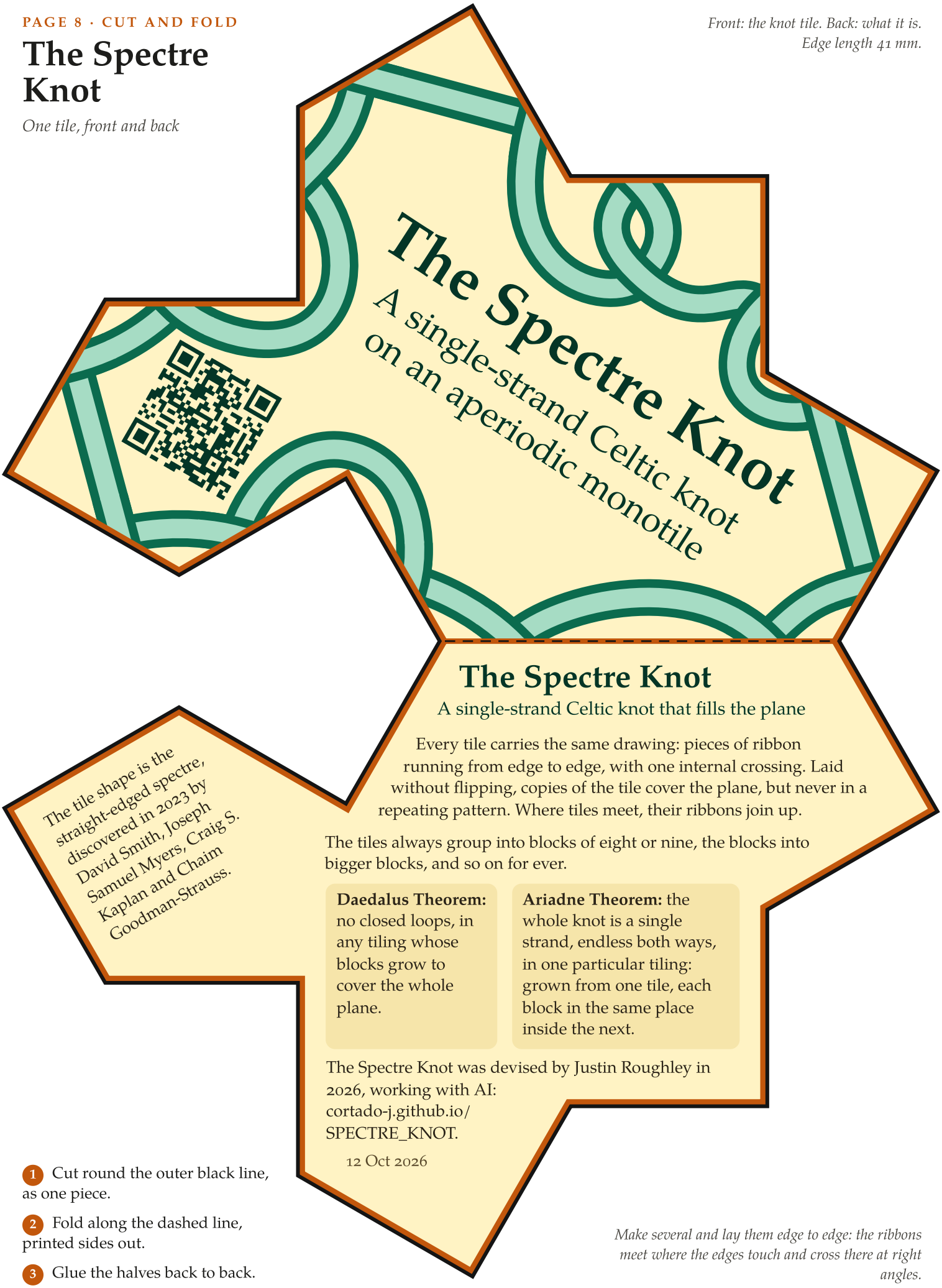
# The Spectre Knot

*One tile, front and back*

*Front: the knot tile. Back: what it is. Edge length 41 mm.*




## The Spectre Knot

A single-strand Celtic knot that fills the plane

Every tile carries the same drawing: pieces of ribbon running from edge to edge, with one internal crossing. Laid without flipping, copies of the tile cover the plane, but never in a repeating pattern. Where tiles meet, their ribbons join up.

The tiles always group into blocks of eight or nine, the blocks into bigger blocks, and so on for ever.

**Daedalus Theorem:** no closed loops, in any tiling whose blocks grow to cover the whole plane.

**Ariadne Theorem:** the whole knot is a single strand, endless both ways, in one particular tiling: grown from one tile, each block in the same place inside the next.

The tile shape is the straight-edged spectre, discovered in 2023 by David Smith, Joseph Samuel Myers, Craig S. Kaplan and Chaim Goodman-Strauss.

The Spectre Knot was devised by Justin Roughley in 2026, working with AI: cortado-j.github.io/SPECTRE_KNOT.

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1. Cut round the outer black line, as one piece.
2. Fold along the dashed line, printed sides out.
3. Glue the halves back to back.

*Make several and lay them edge to edge: the ribbons meet where the edges touch and cross there at right angles.*